

Multiple Choice (10 marks)

- Convert the Fourier series $f(t) = \sum_{k=-\infty}^{+\infty} \frac{1}{1+jk} e^{jkt}$ to trigonometric form.
 - $f(t) = 1 + 2 \sum_{k=1}^{\infty} \frac{\cos kt}{1+k^2}$
 - $f(t) = \sum_{k=-\infty}^{\infty} \frac{\cos kt + k \sin kt}{1+k^2}$
 - $f(t) = 2 \sum_{k=1}^{\infty} \frac{\cos kt + k \sin kt}{1+k^2}$
 - $f(t) = 1 + 2 \sum_{k=1}^{\infty} \frac{\cos kt + k \sin kt}{1+k^2}$
 - $f(t) = 2 + 2 \sum_{k=1}^{\infty} \frac{\cos kt}{1+k^2}$
- The inductance of linear time-varying inductor is given by $L(t) = L_0(t + \tanh t)$ and the current through it is given by $i(t) = \cos t$. Find the voltage across this inductor.
 - $U(t) = L_0 (1 - \tanh t) \cos t + L_0 (t - \operatorname{sech}^2 t) \sin t$
 - $U(t) = L_0 (1 + \operatorname{sech}^2 t) \cos t + L_0 (t + \tanh t) \sin t$
 - $U(t) = L_0 (1 + \operatorname{sech}^2 t) \cos t - L_0 (t + \tanh t) \sin t$
 - $U(t) = L_0 (1 - \operatorname{sech}^2 t) \cos t - L_0 (t + \tanh t) \sin t$
 - $U(t) = -L_0 (1 + \operatorname{sech}^2 t) \sin t + L_0 (t + \tanh t) \cos t$
- Compute the lowest natural frequency for an automobile engine spring. The spring is made of spring steel with 10 active coils, mean diameter of 3.5 in. and wire diameter of 0.20 in.
 - 15 cycles/sec
 - 33 cycles/sec
 - 22.5 cycles/sec
 - 25.8 cycles/sec
 - 18 cycles/sec
- Calculate the heat added when one mole of carbon dioxide is heated at constant volume from 540 to 3540 F.
 - 28,500 Btu
 - 34,650 Btu
 - 40,000 Btu
 - 25,000 Btu
 - 45,000 Btu

5. Both the ALU and control section of CPU employ which special purpose storage location?
 - (a) Multiplexers
 - (b) Memory Cache
 - (c) Registers
 - (d) Flip-Flops
 - (e) Buffers

True / False (10 marks)

Answer True or False

6. True or False: A DC machine commonly has both high shunt and armature resistance values to improve efficiency.
7. True or False: The Laplace transform of $f(t) = \sin(kt)$ is given by $L\{\sin(kt)\} = k / (s^2 - k^2)$ for $t > 0$.
8. True or False: The time-domain response $v_{ab}(t)$ for the given transform is $-48 e^{-3t} - 34e^{-2t} + 16.5e^{-4t}$.
8. True or False: In the decimal number system, the most significant digit (MSD) is the last digit from left to right.
9. True or False: All oscillators listed, including the Hartley oscillator, produce sinusoidal waveforms under ideal conditions.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Calculate and discuss the diffusion rate of water into dry air at 1 atm and 25°C from a test tube with diameter 1 cm.
11. Discuss the process of determining the homogeneous solution for the voltage, $v_C(t)$, across a $(4/3)F$ capacitor in a series RLC circuit with given parameters.

End of Paper